# UNIT-1(Features and nodes of distributed system)

## Features of distributed system

## LECTURE NO:-1

## Distributed system :

Definition and Features of DS : A distributed system consists of a collection of independent autonomous computers, Connected through our network and distribution middleware, Which enables computers to coordinate their activities and to share the resources of the system, so that users perceives the system as a single integrated computing facility .

Or

A distributed system is a collection of independent computers that appears to its users as a single coherent system.

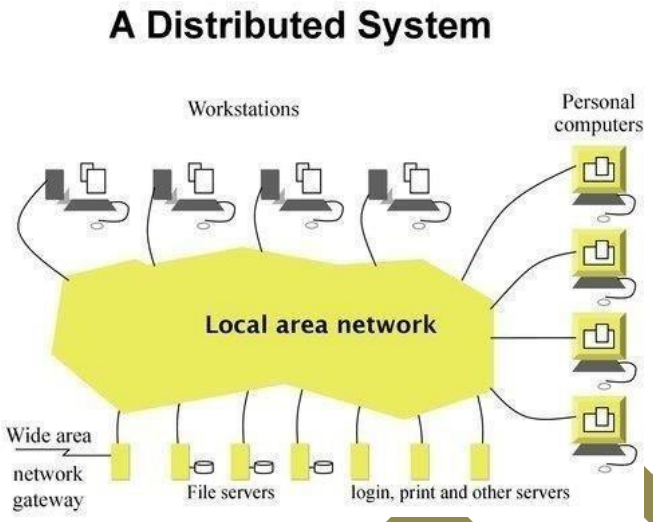
Or

A distributed system is one in which components located at networked computers communicate and coordinate their actions only by passing messages.

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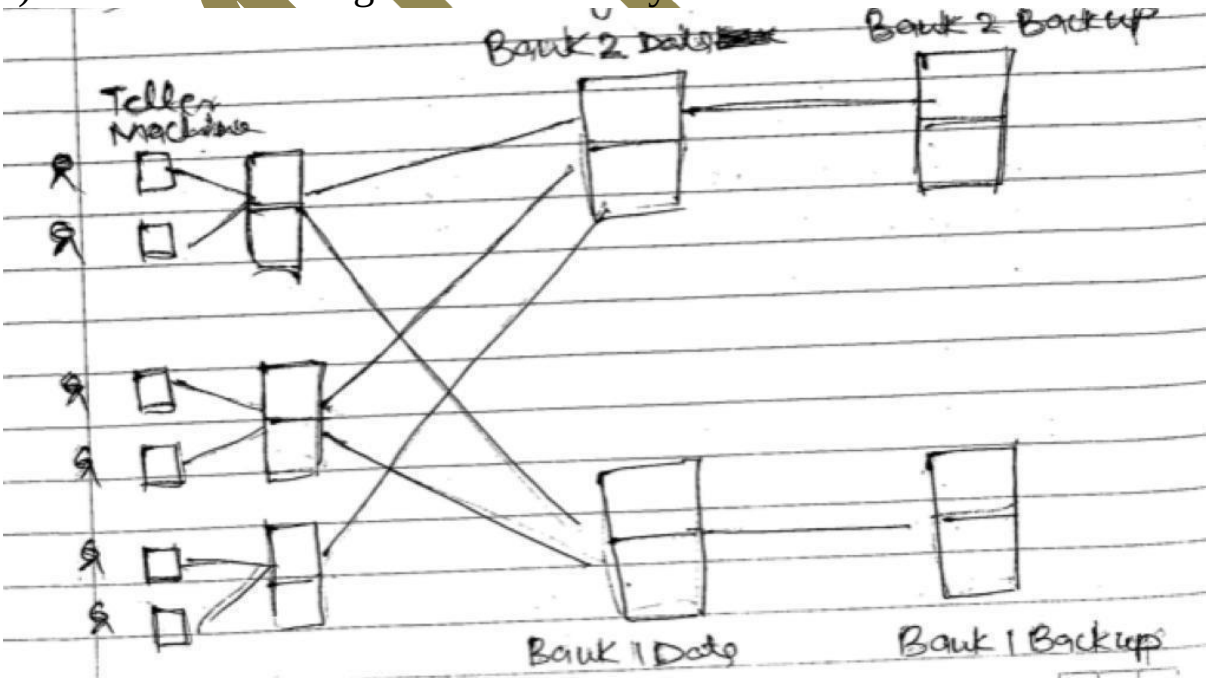
Following are the example of distributed system

**a) Network of workstations**




b) Internet, Intranet, Mobile and ubiquitous computing

c) Automatic banking teller machine system



## Features of distributed system:

1. **Concurrency** : Distributed system working enhances the concurrent program execution in network environment. Components of distributed system, when try to access the shared resources at same time, leads to concurrency, hence it optimizes the performance of the network.
2. **Heterogeneity** : The Internet enables users to access services and run applications over a heterogeneous collection of computers and networks. The heterogeneity applies to all the following
   - Network
   - Computer hardware
   - Operating system
   - Programming language
   - Implementations by different developers
   - Middleware as a software layer helps to achieve heterogeneity.

3. **Openness**: The openness of a computer system is the characteristic that determines whether the system can be extended And re implemented in various ways. The openness of distributed system is determined primarily by the degree to which new resource sharing services can be added.

4. **Security**: Many of the info. Resources that are made available and maintained in distributed system have a high intrinsic value to their users. Security for information resources has 3 components.

   a) Confidentiality: protection against disclosure to unauthorized individuals.
   b) Integrity: protection against alteration or corruption.

c) Availability: protection against interference with means to access the resources.

5. **Scalability** : It is the adaptation of distributed system to accommodate more users , respond faster, it is usually done by adding more and faster resources the components should be designed such that no component should be changed when scale of a system increases.

6. **Failure Handling** : failure in distributed system are partial, that is some components fail while others continue to function. Distributed system must maintain availability even at low level of hardware /software. Fault tolerance is achieved by :

   a) **Recovery**: it involves the design of software so that the state of permanent dates can be recovered and rolled back after a server has crashed.
   b) **Redundancy**: Services can be made to tolerate failure by the use of redundant components example: Two routes in between two routers, database replication in several servers.
   Distributed system provides a high degree of availability in the face of hardware faults.

7. **Transparency**: it is defined as the concealment (The action of hiding something) From the user and the application programmer of the separation of components in a distributed system . so that the system is perceived as a whole rather than a collection of independent components. There are 8 forms of transparency.
   - **Access transparency**: it enables local and remote resources to be accessed using identical operations .
   - **Location transparency**: it enables resources to be accessed without knowledge of their physical or network location .

- **Replication transparency**: it enables multiple instances of resources to be used to increase reliability and performance without knowledge of replicas by users or application programmers.
- **Concurrency transparency**: it enables several process to operate concurrently using shared resources without interference between them.
- **Failure transparency**
- **Mobility transparency** : movement of resources without affecting the operations of others.
- **Performance transparency:** It allows the system to be reconfigured to improve performance as load very.
- **scaling transparency**

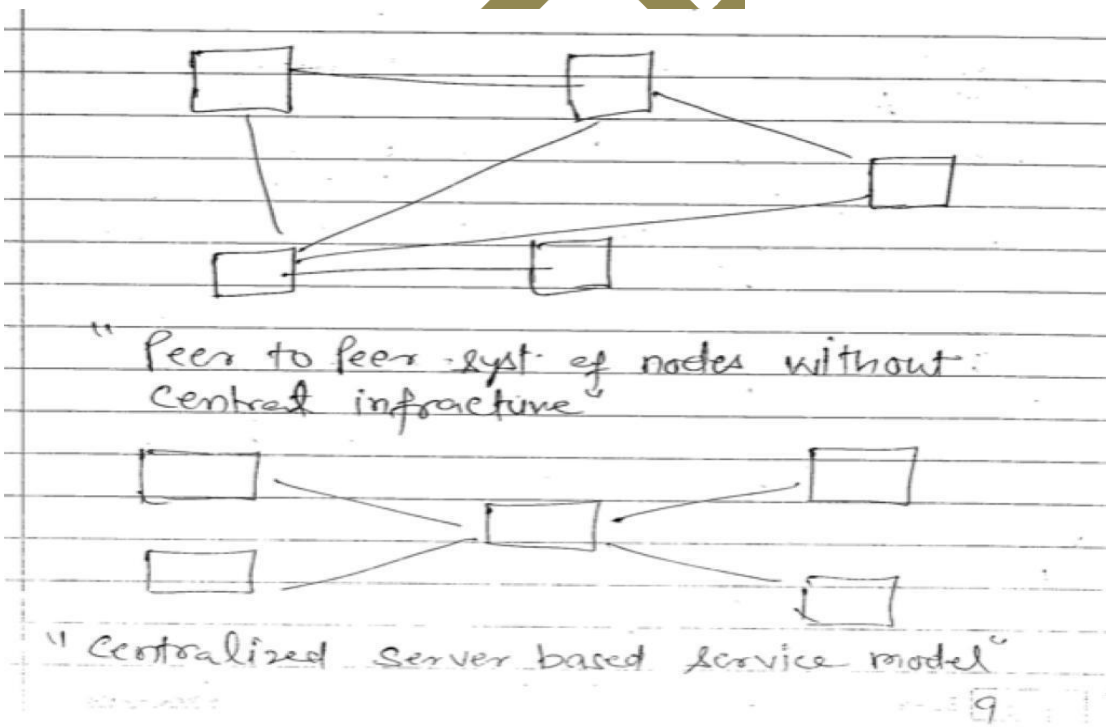
**Nodes of distributed system:**

In communication networks a connection point , or a communication end point. The definition of a node depends on the network and protocol layer referred to a physical network. Node is an active electronic device that is attached to a network and is capable of sending, receiving, forwarding information over a communication channels. In our distributed system the nodes are client, servers and peers, which are as follows

1. **Clients**: The term was first applied to devices, which were not capable of running their stand alone programs , but could interact with remote computers via network. These dumb terminals were clients of the timesharing mainframe computers the clients are generally classifieds as either
    - fat client → that can operate independently
    - Thin client → depend on host
    - Hybrid client→ It is a mixture of fat and thin clients.

2. **Servers**: it is a computer program running to serve the request of other program , the clients. The server perform some computational task on behalf of client example: database servers, file servers, Mail servers , print servers, web servers.

3. **Peers**: Peer to peer computing or networking is a distributed application architecture that partitions tasks or workloads among peers. Peers are both suppliers and consumers of resources.



## UNIT-1(Distributed computing paradigms)

### Distributed computing paradigms

### LECTURE NO:-2

# Distributed computing paradigm , Models of DS :

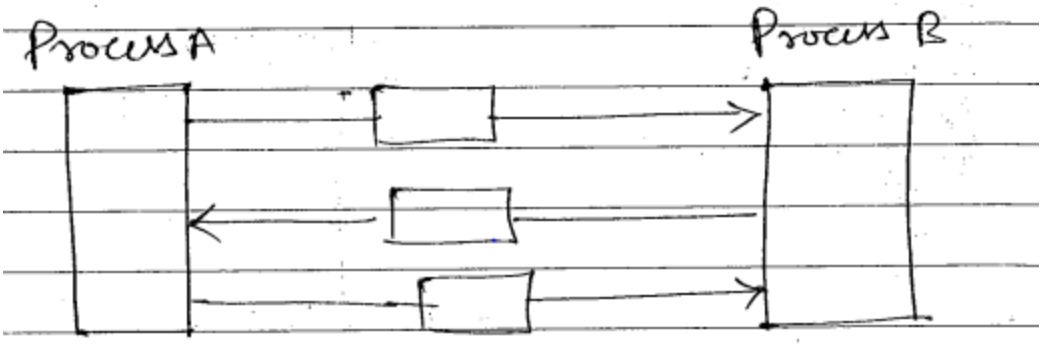
A distributed system consist of multiple autonomous computers that communicate through a computer network. The computers interact with each other in order to achieve a common goal.

Distributed computing also refers to the use of distributed system to solve computational Problems.

**Paradigm for distributed computing:**

It is useful to identify the basic patterns or models of distributed application and classify the detail according these models.

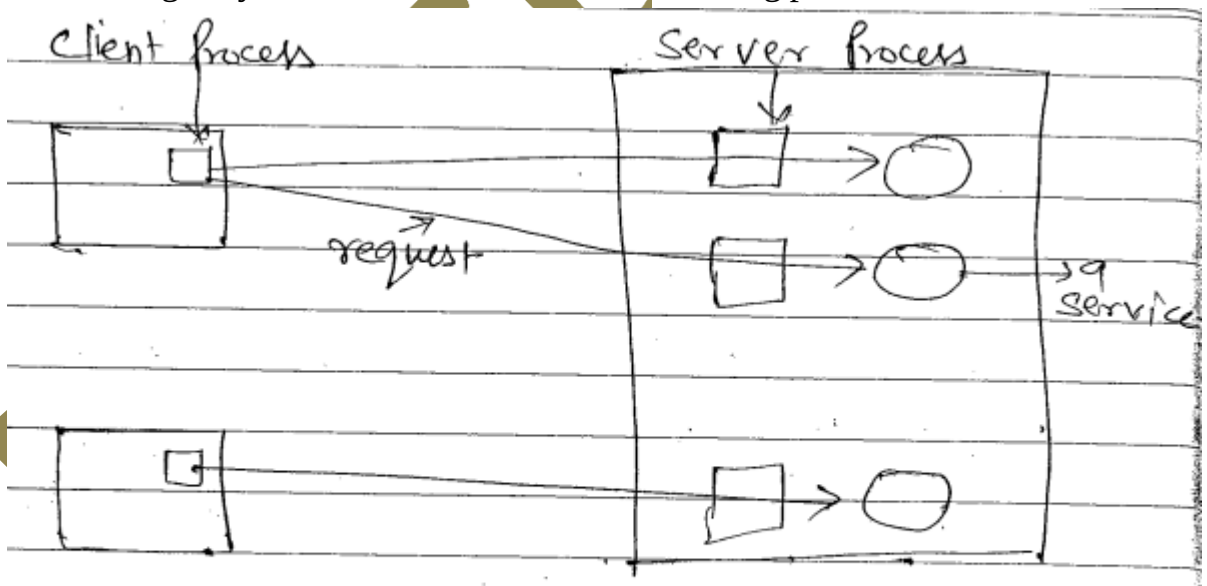
1. **The message passing paradigm:**
   - It is the most fundamental paradigm for distributed applications.
   - A process sends a message representing a request .
   - The message is delivered to a receiver, which process the request and send message in response.
   - In turn, the reply may trigger off further request, which leads to a subsequent reply and so on.




"Message Passing".

2. **The client server paradigm:**

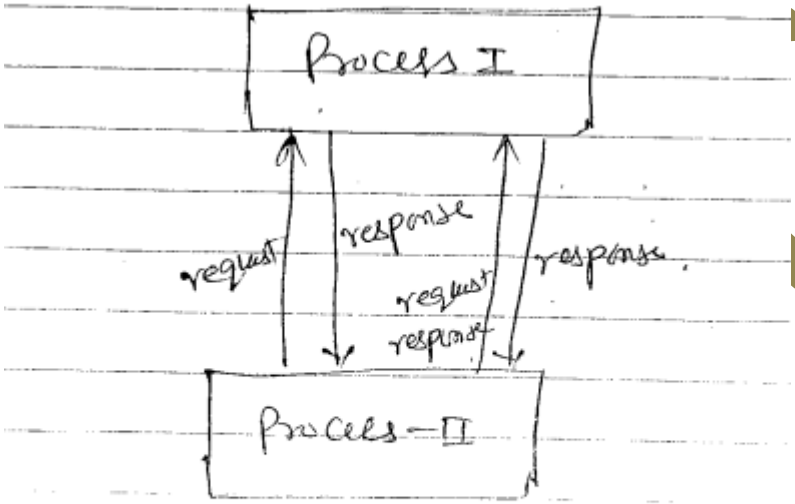
It is the best known paradigm for network applications. the client server model assign asymmetric roles to two collaborating processes.




"The client Server Paradigm"

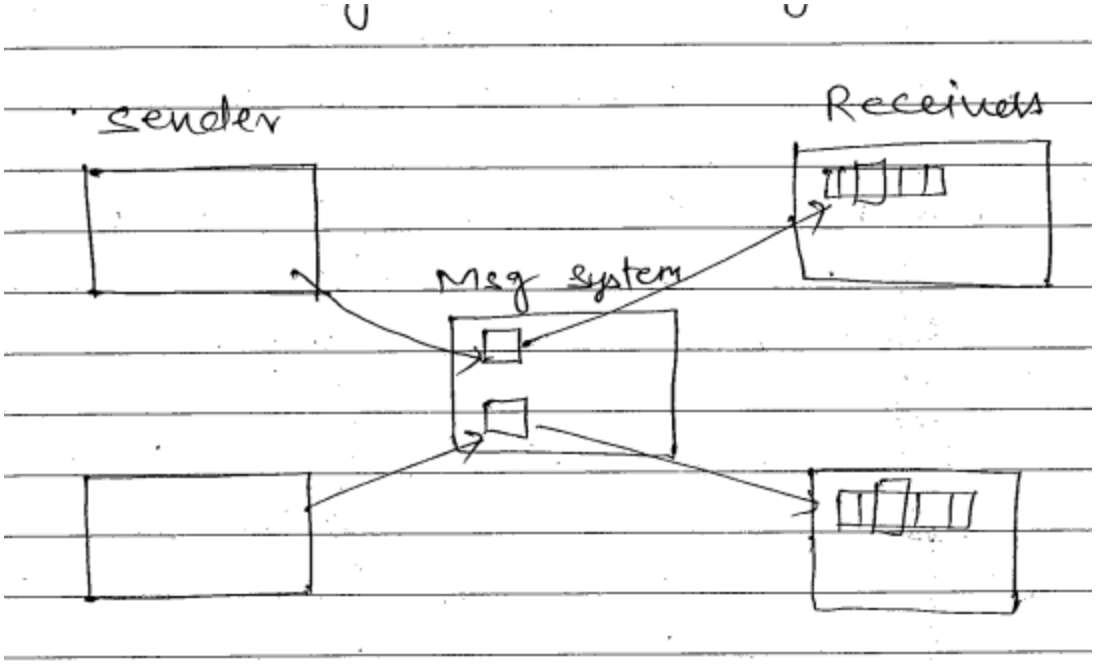
**Peer to peer system architecture:**

In such an architecture the computers that have traditionally been used solely as clients communicate directly among themselves and act as both client and servers, assuming whatever role is most efficient for the network




Peer to peer computing

### 3. The message System paradigm:




A message system or message oriented middleware(MOM) paradigm is an elaboration of basic message passing paradigm.

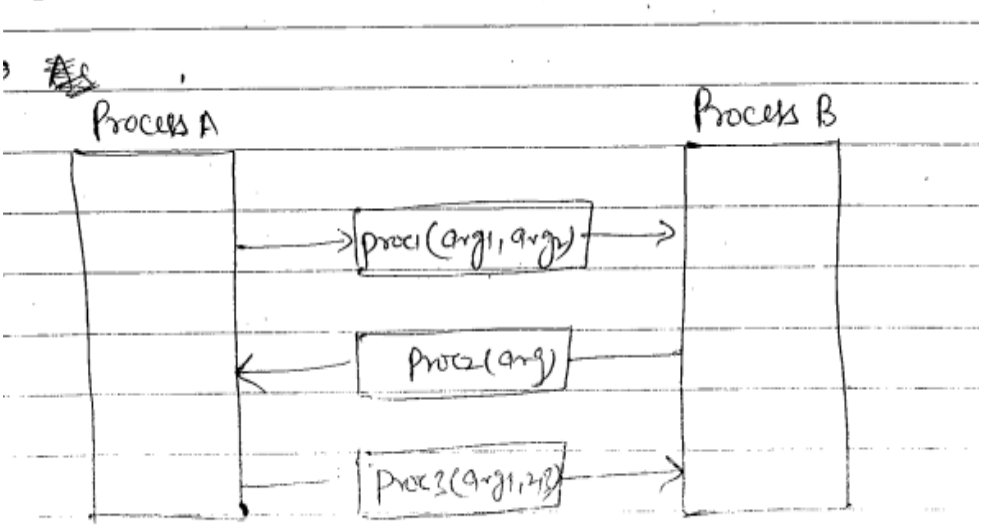
In this paradigm a message system serve as an intermediary among separate independent processes. The message system act as a switch for messages, through which processes exchange message asynchronously in a coupled manner .

2 subtypes of message system model:

a) **The Point to point message model:** In this model a message system forward the message from the sender to the receivers message queue, as discussed above.

**b) The public / subscribe message model**

## 4. Remote procedure call:




- As applications grew increasingly complex, it desirable to have a paradigm which allows a distributed software to program in a manner similar to conventional applications, which run on a single processor.
- The RPC model provides such an abstraction using this model, inter process communication proceed as procedure or function call, which are familiar to application programmer.

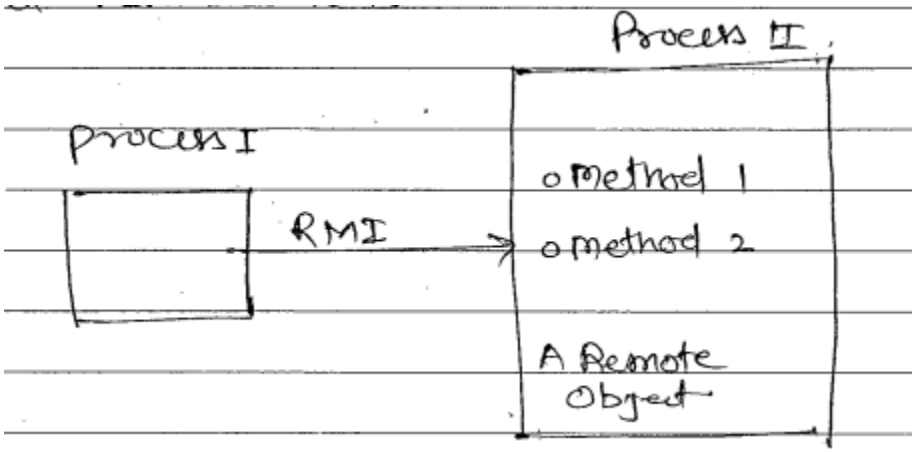
## 5. The distributed object paradigm :

The idea of applying object orientation to distributed application is a natural extension of object oriented software development.

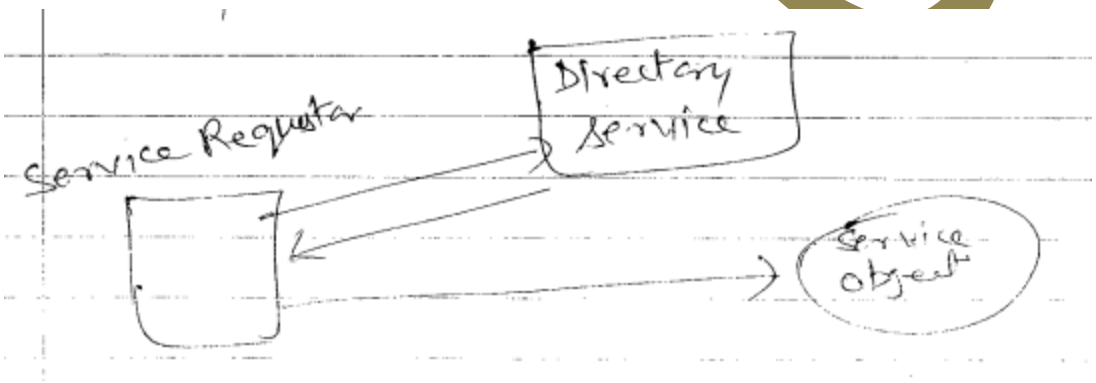
- Object oriented paradigm include:
  a) Remote method invocation
  b) Network services
  c) Object request broker
  d) Objet spaces

a) Remote method invocation:

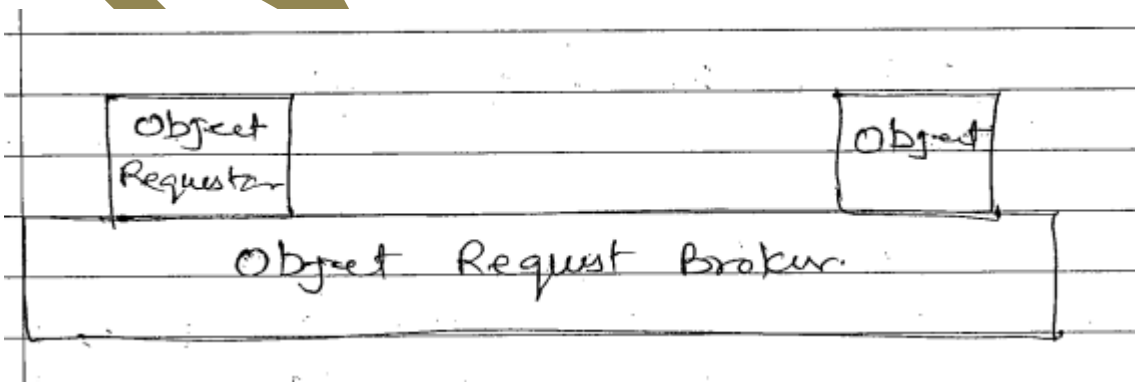
- It is an object oriented equivalent Of RPC . In this model a process invokes the methods in an object, which may reside in a remote host




b) The network service paradigm:
In this paradigm, service providers register themselves with directory servers on a network.

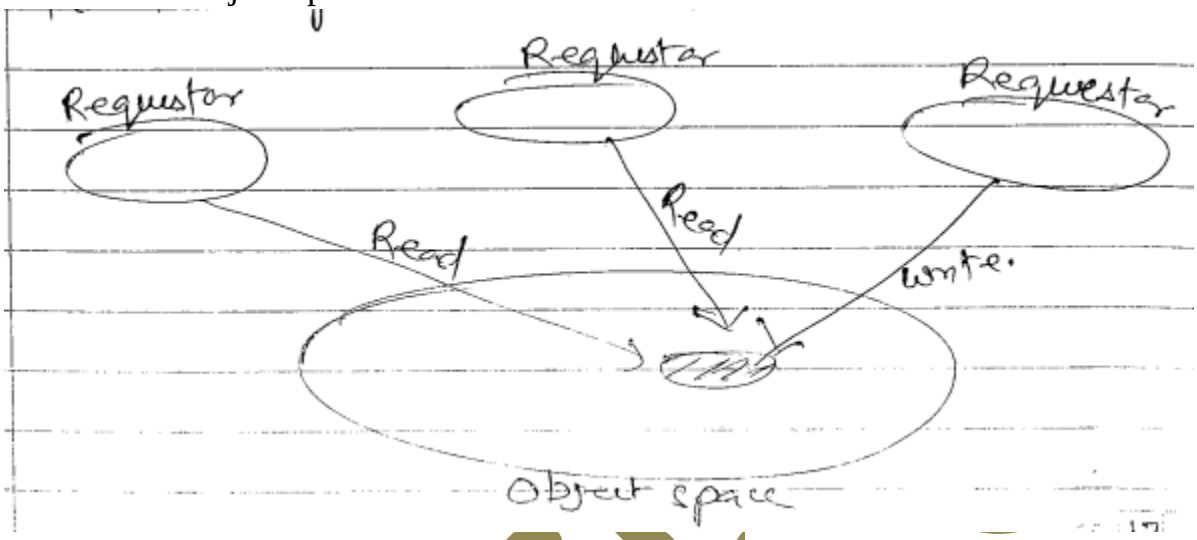



c) The object request broker paradigm :
In this paradigm, an application issues a request to an object request broker(ORB), which directs the request to an appropriate object that provides the desired service.

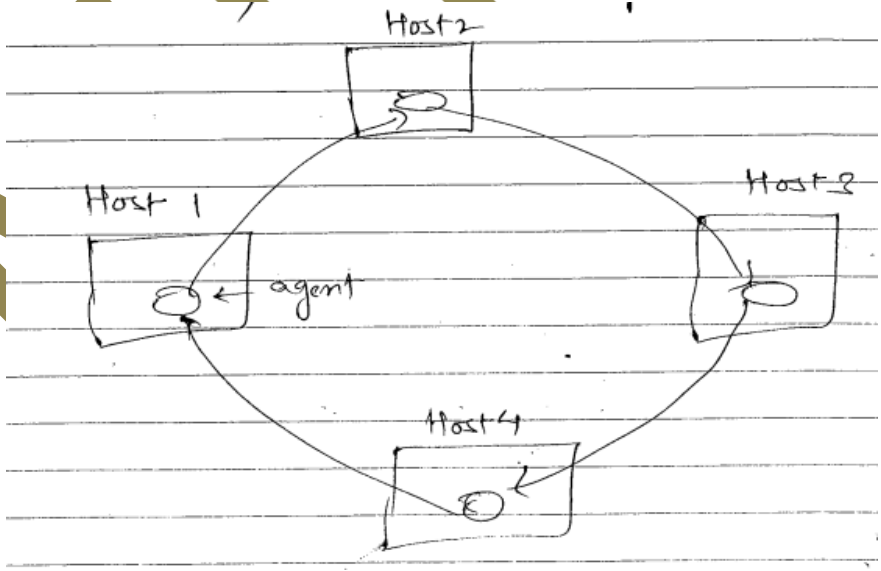


d) Object space paradigm:

- This paradigm assume the existence of logical entities known as object spaces




6. The mobile agent paradigm:
   1. The mobile agent is a transportable program or object
   2. In this model an agent is launched from an originating host
   3. The agent travels from host to host according to an itinerary that it carries



# UNIT-1(Models of distributed system)

## Models of distributed system

## LECTURE NO:-3

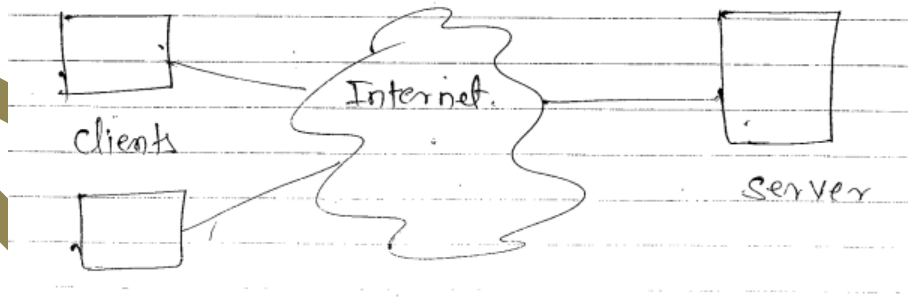
**TOPIC: Models of distributed system:**

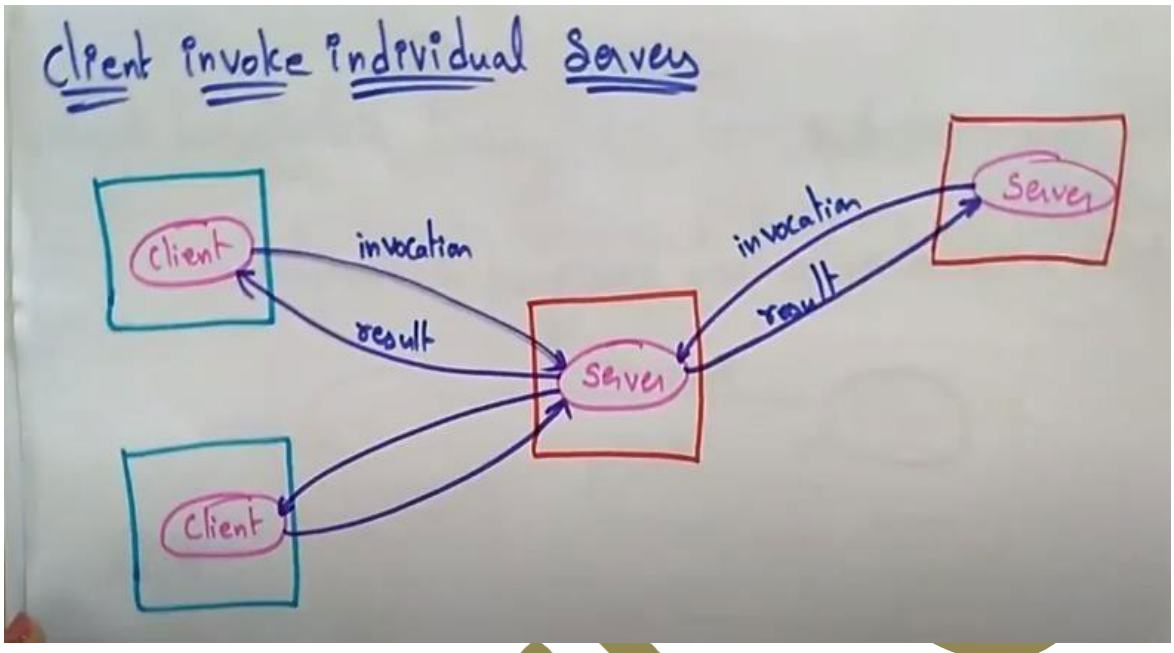
Model is intended to provide an abstract, simplified but consistence desired description of relevant aspect of distributed system design

there are 3 models used in add distributed system .

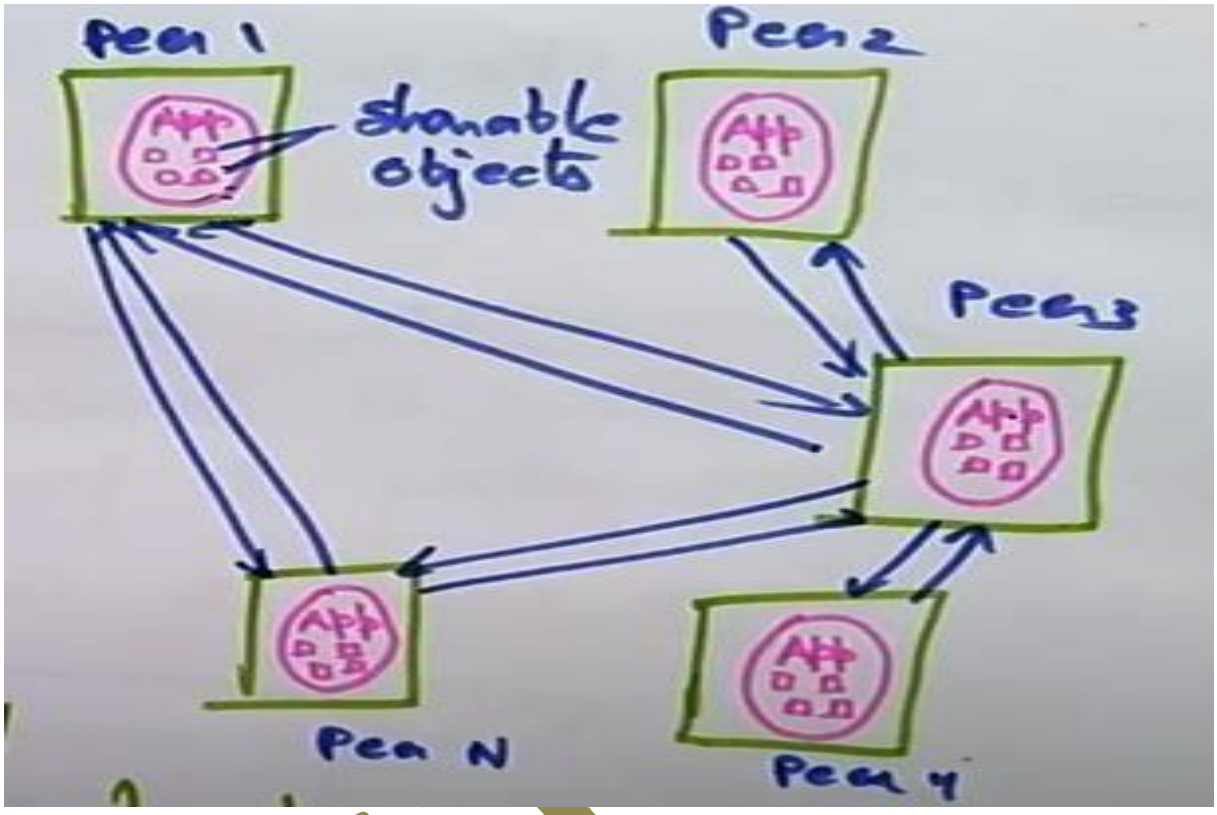
1) **Architectural models :** this model defines the way in which the components of system interact with one another and the way in which they are mapped onto an underline network of computers . Following models are described by architectural model:

   a) Client server model: this model of computing is a distributed application that partitions tasks or workloads between the providers of resources or services, called servers and service Requestors are called clients.

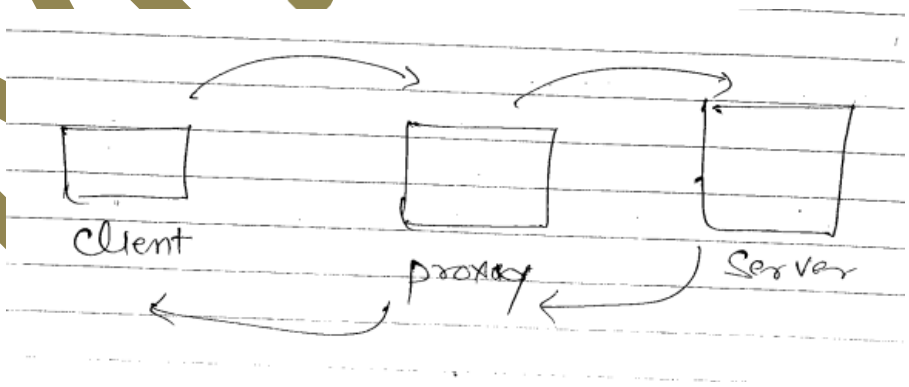





b) Peer to peer model:
In this model computer resources and services are directly exchanged between computer systems.




Proxy servers:
It is a server that acts as an intermediary for requests from clients seeking resources from other servers




Thin client : Also known as lean or slim client .
The thin client is heavily depends on some other computer to fulfil its traditional computational roles

- mobile code
- mobile agent

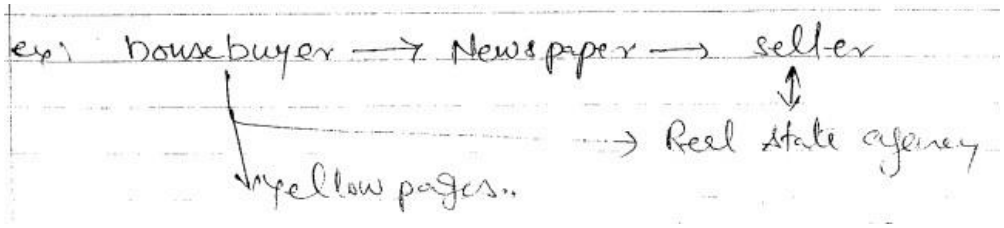
2) Interaction model:
this model deals with performance and with difficulty of setting time limits in our distributed system.

3) Failure model:
This model attempt to give a precise specification of the faults that can be exhibited by processes and communication channel it define reliable communication and correct process following are 4 possible classifications of failures.
   a) omission failure: a component that does not respond to an input from another component and not produce expected output.
      - Process omission failure: The process crashes.
      - Communication omission failure:
      - Network transmission failure

   b) Value failure: a fault that causes our component to respond within the correct time interval but with an incorrect value.
   c) Timing failure : it causes the component to respond with the correct value but outside the specified time of interval
   d) Arbitrary failure the previous failure classes have specified how a component can be considered to fail in either the value or time domain.
      It is possible for a component to fail in both the domains it is called arbitrary failure or Byzantine failure.

4) Cooperative autonomous system: it is a high level service oriented software system that requires the support of communication mechanism on which higher level communication protocols are build in this system each user or process operates autonomously by exporting and requesting services. The key concept in cooperative

autonomous system is integration of services to form cooperative activities.




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# UNIT-1(TYPES OF OS)

## TYPES OF OS

## LECTURE NO:-4

# TYPES OF OS :

Topic: types of operating system.

An operating system is a set of programs that manages computer hardware resources and provide common services for application software.

Following are 4 types of operating system:

a) Centralised operating system :

- These are the traditional operating system for single / multiprocessor architecture.
- The structure of centralised operating system has evolved from a simple control program, responsible for providing a user interface and the control of input output , multitasking system that requires very complex management for processes, memory, files and devices.

The centralised operating system can be viewed as our resource manager, which can be divided into 4 categories:

a) Process management

b) memory management

c) device management

d) file management

2) Network operating system:

- It is also referred to as the dialoguer, is the software that runs on our server and enables the server to manage data, users, groups, security, applications and other networking functions.

- Example: Microsoft Windows server 2003, 2008, Unix, Linux.

Characteristics:

- The network operating system are based on client server architecture in which a server enables multiple clients to share resources

The network OS can do the followings :

- Secure access to network
- Backup data and make sure it always available
- Insure efficient use of our servers capabilities.

3) Distributed operating system:

- It is the logical aggregation of operating system software over a collection of independent, network, communicating nodes.

- The distributed OS is an integral part of the distributed system.

- It provides only the essential services and minimal functionality required of an operating system.

- Although it is distributed in nature, it supports a single system image through the implementation of transparency.

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Notes By: - sanjay pathak(Assistant professor, CSE)

# UNIT-1(Design issues in distributed operating systems)

## Design issues in distributed OS

## LECTURE NO:-5

### Topic: Design issues in distributed OS:

A distributed system from list of concurrent processes, accessing distributed resources (which may be shared or replicated) through message passing in our network environment that may be unreliable and contain UN trusted components.

Following are the design and implementation issues.

Object model and naming services:

- Object in a computer system are processes , data files, memory, devices, processes , networks. It is tempting to assume that all objects can be represented in a uniform way.
- Objector and capsulated in servers and the only visible entity in the system are servers.
- The servers must be identifiable.
- Identification by name
- Identification by service
- Identification by physical or logical address.

- Distributed coordination :
    - The task of coordinator is complicated by the fact of absence of global state. Lack of global state can be today with agreement protocols
    - Barrier synchronisation
    - Condition coordination

- Mutual exclusion
- Deadlock detection/ prevention

Compatibility: it refers to interoperability among resources, that is a resource can be used from any computer or it can be used in combination with other resources. There are 3 levels of compatibility

- Binary level
- Execution level
- Protocol level

d) Performance: performance is arguable the quintessential computing concern and in the distributed operating system, it is no different . Many benchmarks matrix are exists for performance . Like throughput, job completion per unit time, system utilisation etc.

e) Interprocess communication

f) Process management

g) Resource management

h) Reliability:

- Issues arising from availability failure or security valuations are considered faults. These are physical or logical defects that can cause errors in the system Chinese

I ) Synchronisation:

- Corporating concurrent processes have an inherent need for synchronization.

  3 basic situations that define the scope of this need.

- One or more processes must synchronise at a given 0.41 or more other processes to continue

- One or more processes must wait for an asynchronous condition in order to continue
- A process must establish mutual exclusion access to a shared resource

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# UNIT-1(system concepts and architecture)

## system concepts and architecture

## LECTURE NO:-6

**Topic: system concepts and architecture**

Goals: the general goals for the designing of the distributed operating system are not different from any operating system both users and system provided are Concern with the issue of efficiency, flexibility, consistency and robustness of the system .

- Efficiency
- Consistency
- Flexibility
- Robustness

Transparency :

- It is the quality of a distributed operating system to be seen and understand a single system image.

It allows a User to accomplish a system related objective with absolute minimal knowledge of the particular internal details related to the objective.

System goals Transparencies

a) Efficiency——->Concurrency parallelism performance

b) Flexibility——-> access location, migration size revision

c) Consistency-—-> access replication performance

d) Robustness——> failure application size revision

1) Location transparency :

- It comprises 2 distinct SAB aspects of transparency column naming transparency, user mobility

2) Access transparency :

It dictates that any differences in methods of accessing any particular system entity either local or remote must be both invisible to and understandable by user .

3) Migration transparency :

Logical resources and physical processes migrated by the system from one location to another in an attempt to maximize vision C, reliability, availability, security or whatever reason should do so automatically controlled solely by the system. Migration transparency dictates that both the need for end execution of any system entity migration must be bold invisible to and understandable by the user.

4) Replication transparency :

It dictates that both the necessity and execution of application as well as the existenz of replicated entities through out the system must be both invisible and detectable by user

5) Concurrency transparency

6) Failure transparency

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Services:

An operating system is a service provider there are different levels of services the most fundamental service called service primitives are those that must be implemented in the system.

Others can be implemented anywhere in the system and still perform functions basic to the operations of a distributed system , we will call these service provider service servers . Those that performs higher level or special purpose services are value added services.

Architecture model:

Distributed services are described by function Specifications. Their implementation is depend on the system arch Picture and the underline communication network.

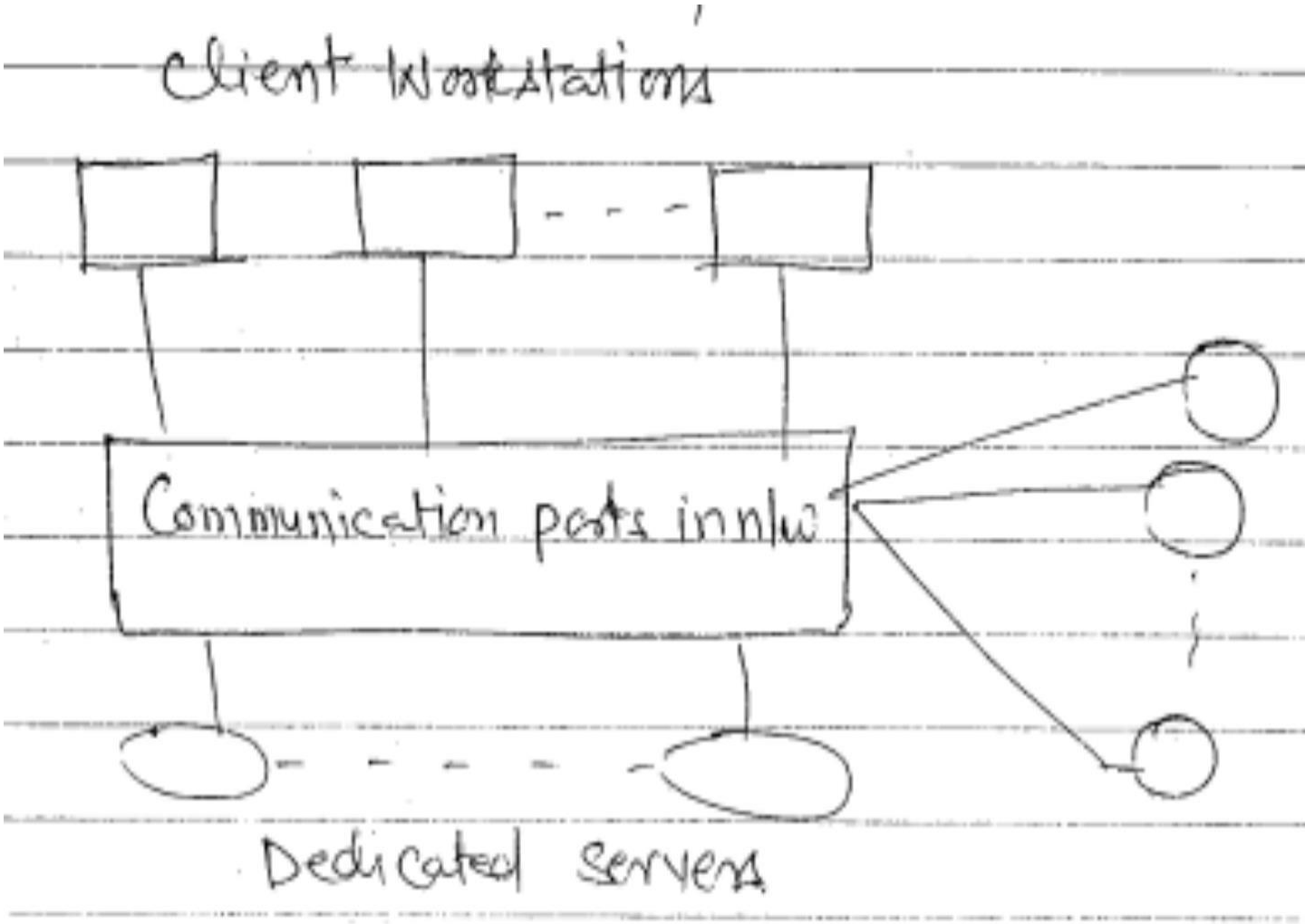
1) Distributed system architecture:

The operating system consist of 2 components processes and files . With the assumption of a network we must add 3rd component the communication path

a) Workstation server model:

b) In this model, workstation is a stand alone computer or as a part of overall network . It provides both local processing capabilities and an interface to the network.

A distributed system based on the workstation server model consist of a few many computers and several workstations interconnected by a communication network.



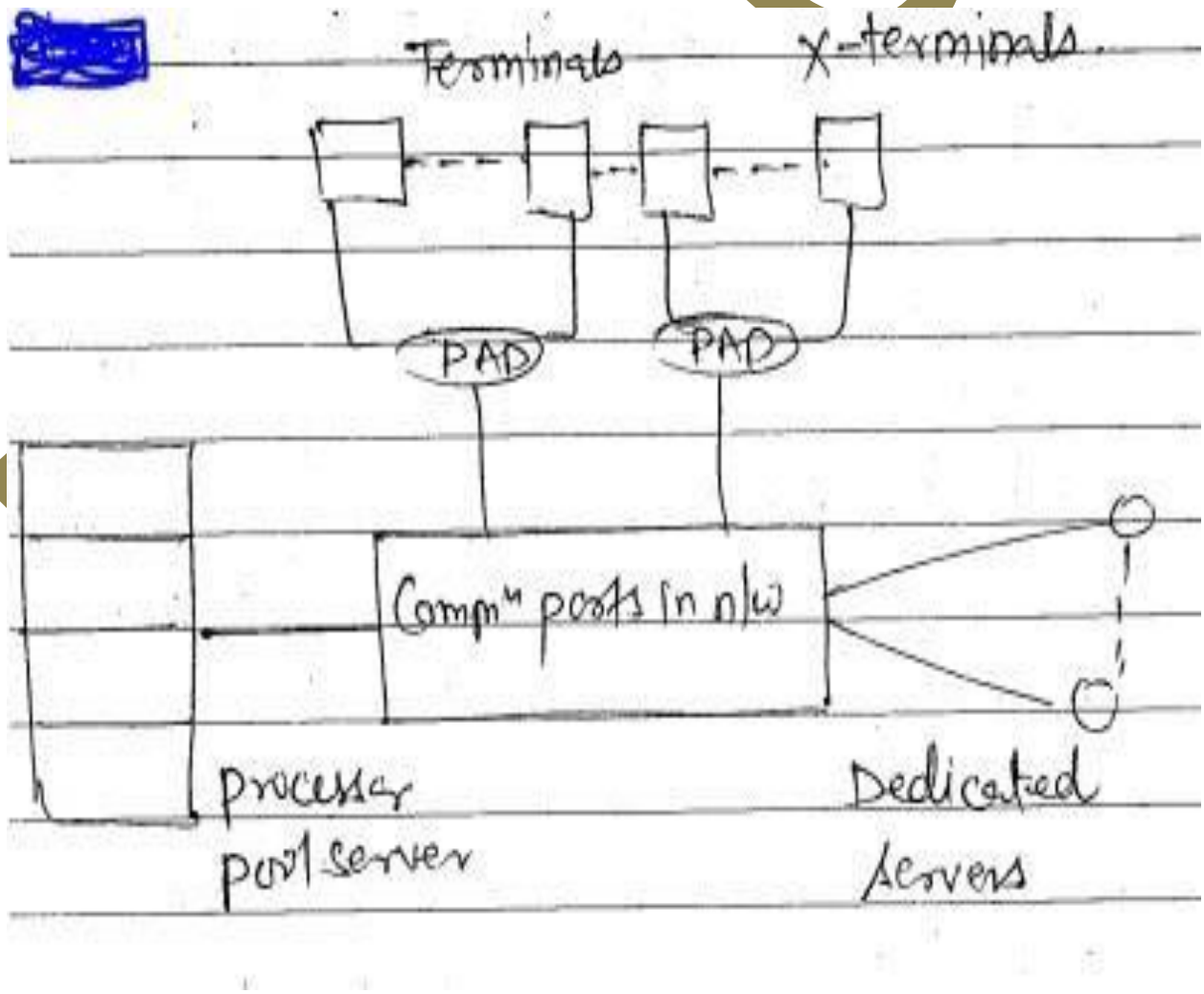

Diskless workstations are also preferred to the disk full workstations from system maintenance point of view. Backup and hardware maintenance are easier to perform with a few large disk then with many small disk scattered all over a building or campus

In the workstation server model, since all files are managed by the file servers. So that users have the flexibility to use any workstation and access to the files in the same manner respective of which workstation the user is currently logged on

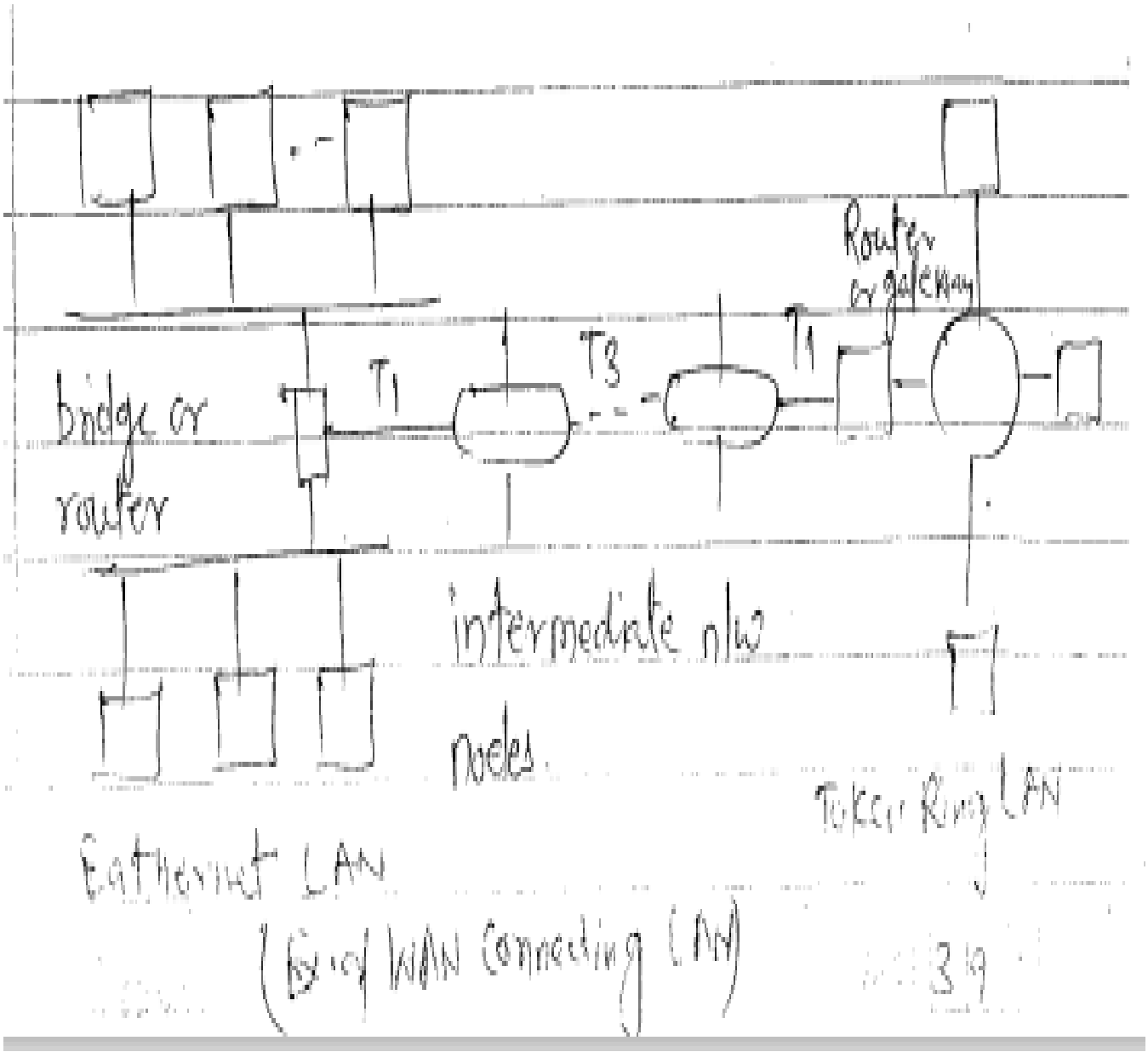
b) Processor pool model:

This model is based on the observation that most of the users does not need any computing power but once in a while he or she may need a very large amount of computing power for a short time.

- the pool of process consist of a large number of microcomputers and minicomputers attached to the network. Each processor in the pool has its own memory tool load and run a system program or an application program of the distributed computing system.



2) Communication network architecture:




Existence of multiple computer systems in distributed system implies the need to interconnect this component.

Hardware interconnection together with the internode.

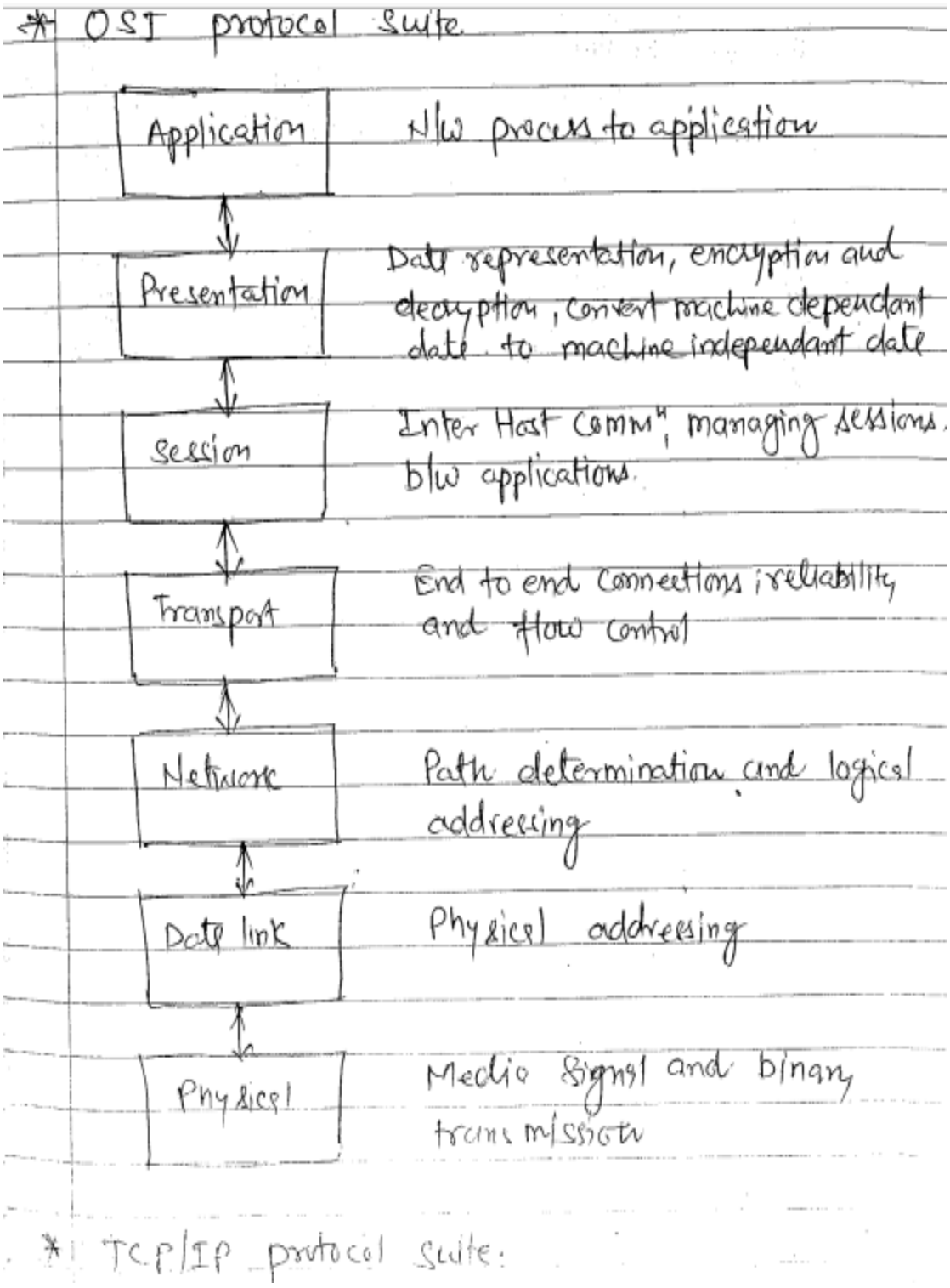
3) Communication network protocol:

- It is a system of digital message for Rules for exchanging those messages in or between computing systems. The protocol main clued signaling authentication and error detection and correction capabilities.
- A protocol define the syntax, semantics and synchronization of communication. In a communication system there are 2 categories of communication services .
  a) Connection oriented communication TCP
  b) Connectionless communication UDP

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OSI Protocol suit

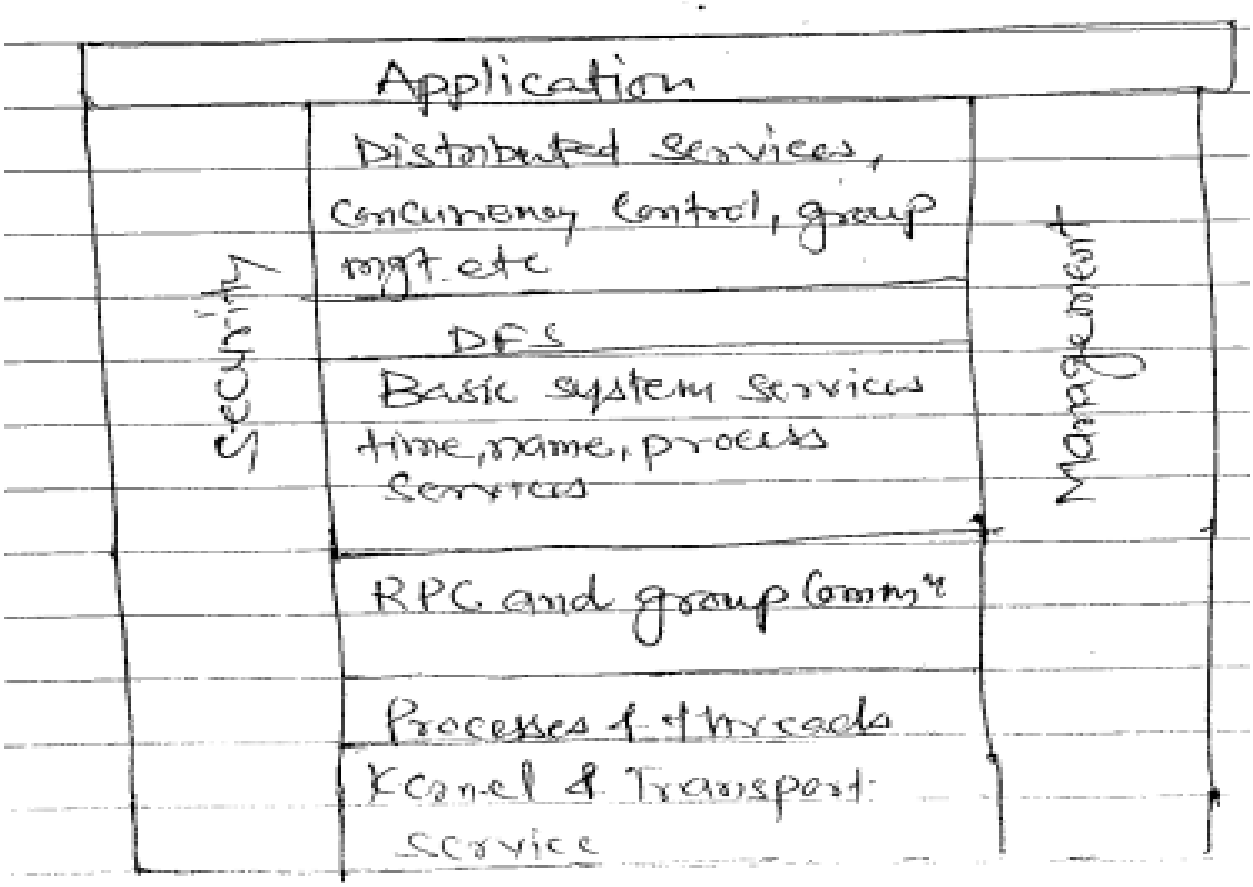
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TCP IP protocol suit

Distributed computing environment:

The dce is a software system developed in the early 1990s.




DCE Architecture

Major components of DCE with every cell are :

1. The security server that is responsible for authentication
2. Sell directory server that is repository of resources and access control list .

3. That distributed time server that provides an accurate Clock for proper functioning of the entire cell . It closely synchronises the clocks of all the computers in the system
4. Thread package
5. RPC: it provides programmer's Vida number of powerful tools necessary to build client server application.
6. Name services : it includes the cell directory services , global directory service and global directory agent. These services allow resources such as servers, files devices and so on to be uniquely named and accessed in a location transparent manner
7. Security Service: it provides the tools needed for authentication and authorization
8. Distributed file server: it provides a file system that has such characteristics as location transparency, high performance and high availability.

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# UNIT-1(Theoretical issues in distributed system)

## Theoretical issues in distributed system

## LECTURE NO:-7

**Topic: Theoretical issues in distributed system**

Notion of state and time:

- Time is the 4th element , it indicates when an event occurred . The state of an entity is the condition or mode. The state of entity depends on its features example the state of a memory cell is the value contained in it.
- If an entity composed of other days, it state contains the state of its component entities.
- The global state of a system is comprised of the state of all entities in the system at a specific instances of time period
- Operating system uses the notions of time and state for performing scheduling of resources and the CPU.

State and events of distributed system:

Local and global states:

- Each component and entity in a system has its own state for example the state of a memory cell is the value contained in it.
- The state of an input output devices indicates whether the device is free or busy and if busy it indicates content of its control register
- We call such state the local state of a component or entity .

- The state of all components and entities in the system at a time instance constitute the state of a system at t.
- the global state does contain the state of all process memories , input output devices and other resources and all communication link in the system . We view it as a collection of local states

Recording the state of a system :

- Information about the state of a system should satisfy 2 conditions correctness and consistency.
- In a conventional system the kernel knows the state of all entities in it's control at any instance of time period hence correctness of state information also implies its self consistency.
- If the state of an application program or a set of application program is desired, the kernel can freeze their execution and record the state of all process involved.
- Consistency means there is no change in database records after the operation apply on the system
- Following algorithms is used to record the global state of the system at any instance of time T .

Algo(naïve state recording)

## Algo (Naive State Recording)

1(a) The initiator node records its own state, the includes the states of all processes and resources located in it.

b) The initiator node sends a message to each neighbouring node asking it to record its state. We will call this a recording message.

2) On receiving a recording message, a node

a) Records the state of the channel over which it received the recording message.

b) If it has not already received a recording message (a node may receive recording messages from many neighbours), the node

(i) Records its own state

(ii) Sends a recording msg to all neighbours except the neighbour from which it received the recording message.

→ Note that a node $N_i$, $i \neq 1$ would receiving a recording message at some time instant $t_i > t$. Hence $N_i$ cannot record its state at time t. In fact all nodes may record their state at different times.

→ Hence the set of these local state does not constitute the global state of the system at time t.

# UNIT-1(clocks and event precedence)

## clocks and event precedence

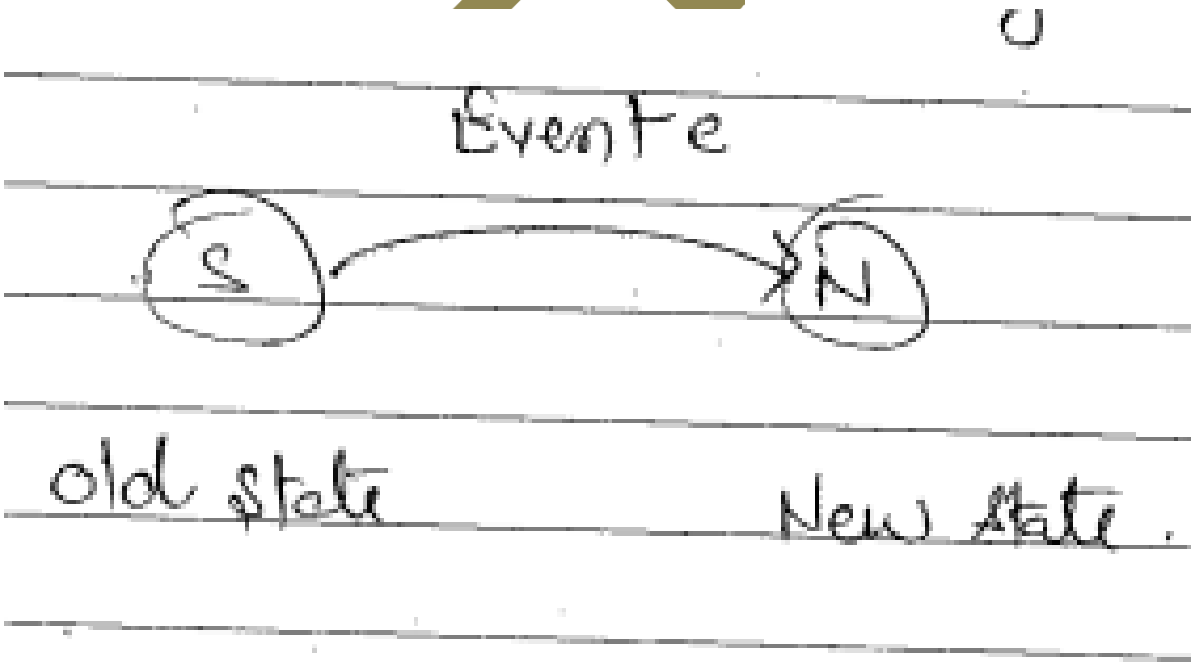
## LECTURE NO:-8

**Topic:** clocks and event precedence

Global state: It is the collection of local states of all entities at the same time stains.

Any collection of local states need not represent consistent global states.

Event column it is represented as follows process ID, old state, new state, event description, channel, message etc .




Time : It is a part of the measuring system used to sequence event to compare the durations of events and the intervals between them. And to quantify rates of change such at the motion of objects.

Clocks an event presidents:

- Turn condition in a distributed system is more complicated than in centralised.
- It is usually not possible to collect all the information about the system in one place, and then let some process examine it and made a decision as is done In the centralised system.
- In general distributed al properties

properties properties.

1. The relevant information is scattered among multiple machine
2. Processes makes decisions based only on local information
3. Single point of failure in the system should be avoided.
4. No common Clock(Global Clock) or other precise global time source

So due to absence of perfectly synchronised Clock and global time in distributed system, the order in which 2 events occur at 2:00 different computers cannot be determined based on the local time at which they occur.

- However under the certain conditions it is possible to a certain the order in which to events based individually on the behaviour exhibited by some computation.

1) Happened before Reaction: It means two events are casually related or not.
   a) a→b if a & b are events in the some process and a occurred before b.

b) a→ b if a is the event of sending a message m in a process and b is the event of receipt of some message by another process.

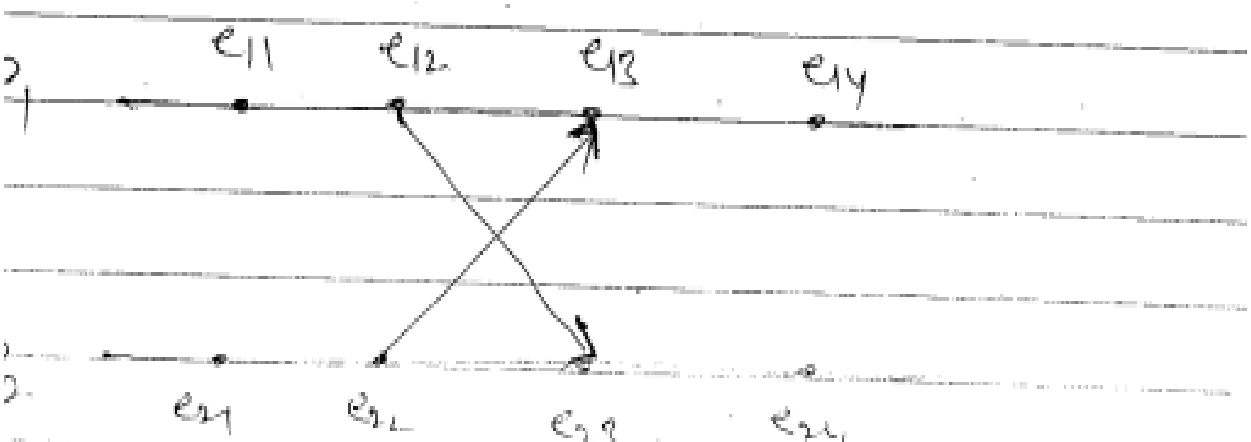
c) A→ , & b→ c then a→c

'→ ' the relation is transitive

- casually related events means put events influence future events. Event a casually affect event by if a arrow be.

2) Concurrent events: Two distinct events a and b are said to be concurrent

(a || b) if $a \not\to b$ & $b \not\to a$. Concurrent events do not casually affect each other.
For any two events a & b in a system, either $a \to b$, $b \to a$ or a || b




A Global Space Time Diagram

In the above space time diagram e11, e12, e13, e14 are events in process p1 and e21, e22, e23, e24 are events in process p2.

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